



Techfiesta **ROBO WORKSHOP**

RULE BOOK

Robo Workshop



DATE 23 - 24 October 2026

VENUE AISSMS Institute of Information Technology, Kennedy Road, Near R.T.O, Pune - 411001, Maharashtra, India

1. Event Date and Timing

Date	23rd to 24th October 2026
Reporting Time	09:30 AM
Workshop Timing	10:00 AM to 04:00 PM

The duration may vary depending on the learning pace and grasping ability of the participants.

2. Venue

AISSMS Institute of Information Technology
Kennedy Road, Near R.T.O., Pune, 411001, Maharashtra, India

3. Contact

Sujal Gaikwad	+91 85309 43237
Parth Kamble	+91 80551 50505

4. Registration Fees

SOLO PARTICIPANT

Rs. 650

per participant

GROUP OF 4

Rs. 600

per participant

TAKE HOME POLICY

Participants registered as a group of 4 may take home the bot they build. Solo participants will be grouped by the organizers and will not take a bot home. See rules 11 to 13.

5. Workshop Overview

This two day hands on workshop takes participants from the fundamentals of electronics and mechanical design through to the assembly and testing of a functional Robo Soccer robot. The learning path moves from concepts and circuit practice to CAD, digital assembly, physical construction, soldering, testing, and final preparation.

CORE LEARNING SEQUENCE

Electronics fundamentals. H-bridge theory. DPDT switch operation. Chassis CAD in Fusion 360. Digital assembly. Physical robot assembly. Soldering connections.

6. Workshop Objectives

By the end of the workshop, participants should be able to:

- Understand basic electrical quantities: voltage, current, and resistance.
- Understand the purpose and working principle of an H-bridge for reversing DC motor direction.
- Use a DPDT switch to manually control motor direction.
- Create a basic robot chassis in Fusion 360 using sheet metal modelling tools.
- Build and inspect a complete digital robot assembly using joints and constraints.

- Identify and assemble the physical chassis, wheels, motors, supports, and related components.
- Follow safe soldering practice and produce reliable electrical connections.

7. Workshop Schedule

7.1. Day 1: Electronics, Motor Control and CAD

Date	23rd October 2026
Morning session	10:00 AM to 12:00 PM
Afternoon session	01:00 PM to 04:00 PM

- **Basic electronics.** Voltage, current, resistance, simple circuit problems, and measurements using a multimeter.
- **H-bridge theory.** Purpose, current direction switching, transistor and MOSFET based circuit concepts, and practical motor testing.
- **DPDT switch.** SPST, SPDT, and DPDT switch types, and using a DPDT switch as a manual H-bridge.
- **CAD of the chassis.** Fusion 360 sketching, Extrude, Projection, Bending, Fillet, and Chamfer operations.
- **Digital robot assembly.** Importing CAD parts, applying joints and constraints, checking interference, and validating movement.

7.2. Day 2: Physical Assembly, Soldering and Final Testing

Date	24th October 2026
Morning session	10:00 AM to 12:00 PM
Afternoon session	01:00 PM to 04:00 PM

- **Physical robot assembly.** Identify chassis components and assemble wheels, motors, supports, and related parts under guidance.
- **Soldering.** Safety practices, correct solder joint technique, motor wire connections, switch connections, and power supply leads.
- **Testing and final preparation.** Inspect the robot, correct loose or unsafe connections, verify movement, and prepare the competition ready soccer bot.

8. Participant Rules and Safety

1. All participants must register and follow the instructions of the workshop coordinators.
2. Follow all safety instructions and never operate equipment, circuits, or robots in an unsafe manner.
3. Disconnect power before changing, rewiring, modifying, or soldering any circuit.
4. Check circuit connections and polarity before powering the system.
5. Handle all tools, components, equipment, and workstations carefully, and return them after use.
6. Keep the work area clean and report any damaged component, loose connection, overheating, or unusual behaviour immediately.
7. Ensure all motors, wheels, wires, and mechanical parts are securely installed and do not interfere with movement.
8. Inspect and test the robot safely before operation, starting with basic power and motor checks.
9. During Robo Soccer, use only approved robots and systems, and do not intentionally interfere

with or damage another robot or the playing field.

10. Follow the instructions of the referee and coordinators at all times. Their decision regarding safety, discipline, and competition conduct is final.
11. Participants registered as a group of 4 may take home the bot they build.
12. Individual participants will not take bots home.
13. Solo participants will be placed into groups decided by the organizers, and the bot built by such a group will not be taken home.

9. Quick Participant Checklist

- Attend the safety briefing.
- Bring or arrange the required workshop materials as instructed.
- Keep a notebook for circuit diagrams, measurements, and design notes.
- Follow the instructor during live circuit demonstrations.
- Save CAD work regularly and keep an organized assembly.
- Check wiring before every power up.
- Use soldering equipment only at the designated workstation.
- Complete the final robot inspection before the competition.

TIP

Bring your own soldering gun if you have one.

10. General Policies

Conduct, equipment, certification, and organizer authority, the essentials in one place.

10.1. Conduct and Fair Play

- Respect fellow participants, mentors, coordinators, referees, and support staff.
- No deliberate damage, tampering, theft, or unauthorized modification of another team's robot or components.
- Teams must solve technical problems through their own work, and may request guidance from instructors where permitted.
- Abusive, disruptive, or unsafe behaviour may lead to removal from a practical session or the competition.
- All teams get equal access to the announced workshop resources, subject to equipment availability.

10.2. Certification and Attendance

CERTIFICATE OF PARTICIPATION

Participants should attend the complete workshop schedule to receive the participation certificate, subject to the organizer's final certification policy. Attendance, discipline, and participation may be recorded by the workshop coordinators.

11. Competition

There will be a mini competition after the workshop to check the effectiveness of the workshop and to test the bots.

12. For More Information

Website ioittenet.com **Instagram** [@ioit_tenet](https://www.instagram.com/ioit_tenet)

This rulebook is subject to final approval by the workshop organizers. Any updates will be communicated to participants before the relevant session.

Where Ideas Become Robots.

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